

主图 ZDIF 线推导过程

函数 EMA 的定义：

$$Y = \text{EMA}(X, N) = \frac{N-1}{N+1} \cdot Y_1 + \frac{2}{N+1} \cdot X$$

指数平滑移动平均线 MACD(12,26)的定义：

$$\begin{cases} DIF = MS - ML = \text{EMA}(C, 12) - \text{EMA}(C, 26) \\ MS = \text{EMA}(C, 12) = \frac{11}{13} \cdot MS_1 + \frac{2}{13} \cdot C = MS_1 + \frac{2}{13} \cdot (C - MS_1) \\ ML = \text{EMA}(C, 26) = \frac{25}{27} \cdot ML_1 + \frac{2}{27} \cdot C = ML_1 + \frac{2}{27} \cdot (C - ML_1) \end{cases}$$
$$\begin{aligned} & DIF - DIF_1 \\ & \equiv (MS - ML) - (MS_1 - ML_1) \\ & \equiv [MS_1 + \frac{2}{13} \cdot (C - MS_1) - ML_1 - \frac{2}{27} \cdot (C - ML_1)] - (MS_1 - ML_1) \\ & \equiv \frac{2}{13} \cdot (C - MS_1) - \frac{2}{27} \cdot (C - ML_1) \\ & \equiv \frac{2}{13 \times 27} \cdot (14 \cdot C + 13 \cdot ML_1 - 27 \cdot MS_1) \end{aligned}$$

当 DIF 转向，即 $DIF - DIF_1 \geq 0$ 。从而，

$$\begin{aligned} C & \geq \frac{1}{14} \cdot (27 \cdot MS_1 - 13 \cdot ML_1) \approx 2 \cdot MS_1 - ML_1 \\ & = MS_1 + (MS_1 - ML_1) \end{aligned}$$